

Detailed Data-Generation Write-up for the EncounterNN Workstream

Overview

The encounterNN data-generation pipeline was built because the earlier scalar Zone-3 dataset was not suitable for training an encounter-level neural surrogate. The scalar Zone-3 dataset contained pair-level features and scalar correction targets, such as:

- pair separation,
- softened distance,
- pair masses,
- timestep,
- normalized radial velocity,
- normalized tangential velocity,
- scalar correction target.

That dataset was appropriate for training a scalar force multiplier, but the encounterNN required a different kind of data: a full 3-body start state, a full 3-body IAS15 exit state, a full 3-body SIMON-noNN exit state, and the residual between them. Therefore, the encounterNN generators created new full-state transition data.

The common encounterNN target used by all three generators is:

$$R = F_{IAS15} - F_{noNN}$$

where F_{IAS15} is the encounter-window exit state from IAS15 and F_{noNN} is the encounter-window exit state from SIMON-noNN. The neural network is trained to predict this residual. At inference, the corrected state is:

$$F_{corr} = F_{noNN,new} + \hat{R}$$

This means the encounterNN is not learning a scalar force correction. It is learning an encounter-window exit-state residual.

1. `generate_encounter_surrogate_data_v1.py`

Purpose

`generate_encounter_surrogate_data_v1.py` is the first full-state encounter-surrogate data generator. Its purpose is to create prototype training shards for an encounter-level residual model at:

- $dt = 0.08$,
- window length = 0.5 years,
- active pair = 1-2 by default,
- Zone-3 start region approximately $0.052 < r < 0.148$ AU.

The file explicitly states that it is **not** generating the old scalar- c correction dataset. Instead, for every accepted Zone-3 encounter state, it builds a full 3-body state, runs IAS15 and SIMON-noNN over the same window, and saves the residual target:

$$residual = IAS15_{final} - noNN_{final}$$

Step-by-step process

Step 1: Set physical constants and zone thresholds

The file defines the revised SIMON constants:

- gravitational constant $G = 1.0$,
- softening $\epsilon = 3 \times 10^{-4}$,
- Zone 2 / adaptive threshold $r < 0.05$ AU,
- Zone 3 upper threshold $r < 0.15$ AU,
- clean Zone-3 sampling range $0.052 < r < 0.148$ AU,
- hard near-singularity gate around 4×10^{-4} AU,
- maximum adaptive substeps = 16.

These constants align the generated data with the revised Phase-2 SIMON methodology: Zone 2 is handled by direct Newtonian force plus adaptive sub-stepping, while Zone 3 is the learned correction region.

Step 2: Try to load the old scalar Zone-3 dataset as a proposal distribution

The file has a command-line argument:

```
--scalar-data encounter_data_zone3_v3_augmented.npz
```

The code attempts to load this file and use its pair-level fields:

```
r_AU  
log_mi  
log_mj  
log_dt  
v_rad_norm  
v_tan_norm
```

However, this is only for sampling guidance. The old scalar dataset is not the training data for encounterNN.

If the scalar file is missing, the generator does not stop. It prints that the scalar-data file was not found and switches to a fallback random sampler. Therefore, v1 was designed to use the old scalar dataset **only if available**; otherwise, it samples from hand-coded fallback ranges.

Step 3: Choose an encounter category

The generator supports several sampling modes:

```
broad
strong_approach
weak_side
recede
mixed
```

For `mixed`, the code deliberately balances the categories:

- 40% strong approach,
- 30% weak/side,
- 30% recede.

The categories are based on normalized radial velocity:

- strong approach: $v_{rad,norm} < -0.6$,
- weak/side: $-0.6 \leq v_{rad,norm} \leq 0.2$,
- recede: $v_{rad,norm} > 0.2$.

This was motivated by the earlier scalar Zone-3 work, where the radial velocity sign was important: approaching and receding encounters require different correction behaviour.

Step 4: Sample pair-level encounter variables

The generator selects:

- active pair separation r_{pair} ,
- log mass of body i ,
- log mass of body j ,
- normalized radial velocity,
- normalized tangential velocity.

If the old scalar dataset is available, these values are sampled from the empirical scalar dataset. If not, the generator samples them from fallback ranges that mimic the Zone-3 distribution.

This is where the old scalar dataset influenced the design: it motivated the variables and realistic ranges. But it did not provide the full encounterNN input state.

Step 5: Construct a full 3-body state

After choosing the active-pair variables, the generator constructs a full planar 3-body state.

It does this by:

1. selecting the active pair, usually bodies 1 and 2,
2. assigning pair masses from sampled log masses,
3. sampling or assigning the third-body mass,
4. choosing a random 2D orientation for the active pair,
5. placing the pair at the required separation,
6. assigning relative radial and tangential velocity to match $v_{rad,norm}$ and $v_{tan,norm}$,
7. placing the third body at a safe distance,

8. shifting the whole system into the centre-of-mass frame.

This is the key difference from the scalar dataset. The output is now a full state, not just a pair feature row.

Step 6: Check that the generated state is valid

The generator checks that:

- all positions and velocities are finite,
- the active pair really starts in Zone 3,
- the system is not already ejected,
- the minimum distance is not invalid,
- energy can be computed.

If a state fails these checks, it is rejected and the rejection reason is counted.

Step 7: Run IAS15 over the encounter window

The generator runs IAS15 for the fixed 0.5-year window. IAS15 is used as the reference solution.

It rejects the sample if:

- IAS15 fails,
- IAS15 ejects,
- IAS15 goes below the minimum allowed distance,
- IAS15 energy drift exceeds the allowed threshold.

The default IAS15 minimum-distance filter keeps the first prototype clean by rejecting windows that go deep into Zone 2.

Step 8: Run SIMON-noNN over the same window

The generator then runs the revised noNN baseline over the same endpoint. In this context, noNN means:

- Zone 2 uses direct Newtonian force plus adaptive sub-stepping,
- Zone 3 uses the same non-NN baseline rule rather than a learned scalar correction,
- Zone 4 uses direct Newtonian force.

It rejects the sample if:

- the noNN integration fails,
- the noNN branch ejects,
- noNN energy drift is too large.

Step 9: Compute the residual target

For accepted samples, it computes:

$$residual_x = x_{IAS15,final} - x_{noNN,final}$$

$$residual_v = v_{IAS15,final} - v_{noNN,final}$$

It also computes position and velocity residual RMS values. Extreme residual outliers can be rejected.

Step 10: Build feature vectors

The file saves two feature representations.

The first is `X_raw21`:

```
[log masses, full positions, full velocities]
```

This is 21 values: 3 log masses + 9 position coordinates + 9 velocity coordinates.

The second is `X_rel18`, a compact relative representation:

```
[log masses,
 pair relative position,
 pair relative velocity,
 third-body position relative to pair COM,
 third-body velocity relative to pair COM,
 log(dt),
 v_rad_norm,
 v_tan_norm]
```

This gives 18 inputs. This became the main model input used by the encounterNN trainer.

Step 11: Save fields to a compressed shard

For every accepted sample, the file saves:

- `m`,
- `x0`,
- `v0`,
- `x_ias_final`,
- `v_ias_final`,
- `x_nonn_final`,
- `v_nonn_final`,
- `residual_x`,
- `residual_v`,
- `X_raw21`,
- `X_rel18`,
- `dt`,
- `window_years`,
- active pair IDs,
- `r_pair`,
- `r_soft_pair`,
- `v_rad_norm`,

- `v_tan_norm`,
- `source_row_idx`,
- `category_id`,
- energy values,
- energy drift values,
- angular momentum,
- minimum distance,
- maximum radius,
- noNN zone counts,
- residual RMS diagnostics.

The output is a `.npz` shard plus a `_summary.txt` file describing pass rate, category counts, energy, residuals, and rejection reasons.

What v1 achieved

v1 established the full-state residual dataset format. It proved that the pipeline could generate physically filtered, full 3-body encounter-window samples. It also showed that a universal correction across all encounter phases was too broad: strong-approach samples were learnable, but weak/side and receding samples were often harmed by applying a residual correction unnecessarily.

2.

`generate_encounter_surrogate_data_v2_event_targeted_fixed.py`

Purpose

`generate_encounter_surrogate_data_v2_event_targeted_fixed.py` extends the v1 generator by adding an event-targeted sampling mode. The purpose was to generate data closer to the real failed rollout event observed at:

- $dt = 0.08$,
- active pair = 1-2,
- event time around $t = 82.56$ years,
- $r_{pair} \approx 0.139$ AU,
- $v_{rad,norm} \approx -1.1$,
- $v_{tan,norm} \approx 0.85$.

The v2 file keeps the same core residual target:

$$residual = IAS15_{final} - noNN_{final}$$

but changes the sampling strategy to better cover the difficult real-event region.

Step-by-step process

Step 1: Keep the same core physics and output format as v1

The v2 generator keeps the same basic workflow as v1:

1. build a full 3-body Zone-3 start state,
2. run IAS15 for 0.5 years,
3. run revised SIMON-noNN for the same window,
4. compute and save the residual target,
5. save the same full-state arrays and diagnostics.

This allowed v2 shards to be merged with v1 shards and used by the same trainer and inspector.

Step 2: Load the old scalar dataset if available

Like v1, v2 attempts to load the old scalar Zone-3 dataset using `--scalar-data`. It uses the old scalar fields only as an empirical proposal distribution for:

- r ,
- log masses,
- $v_{rad,norm}$,
- $v_{tan,norm}$.

If the file is missing or does not contain the required fields, v2 also falls back to random sampling.

Step 3: Add `event_strong_approach` mode

The main new feature in v2 is the mode:

```
event_strong_approach
```

This was added because the earlier empirical proposal did not sufficiently cover the exact real rollout event. The code explicitly notes that the original $dt=0.08$ proposal under-covered the region around:

```
v_rad_norm ≈ -1.1  
v_tan_norm ≈ 0.85  
r_pair ≈ 0.139 AU
```

Therefore, v2 does not rely only on empirical rows for this region. It uses explicit targeted sampling ranges to enrich that event-like part of the state space.

Step 4: Select scalar proposal rows or targeted fallback values

For ordinary modes, v2 behaves like v1 and uses masks such as:

- `strong_approach`,
- `weak_side`,
- `recede`,

- `broad`.

For `event_strong_approach`, it first looks for scalar rows satisfying approximately:

```
v_rad_norm < -0.85
0.60 <= v_tan_norm <= 1.20
0.095 <= r_pair <= 0.148
```

But the code also explicitly samples the event-like region, because the real event was not well covered by the original proposal.

Step 5: Construct the full 3-body state

The full-state construction remains conceptually the same as v1:

- choose the active pair,
- set pair masses,
- create the pair separation,
- create radial/tangential velocity consistent with the selected features,
- place the third body,
- centre the system in the centre-of-mass frame.

The difference is that in `event_strong_approach` mode, the active pair and masses are made more IC1-like:

- active pair 1-2,
- pair masses around 0.01 and 0.005,
- third body around mass 1.0.

This better matches the actual IC1 event geometry.

Step 6: Run the same IAS15 and noNN windows

For each candidate, v2 still runs:

- IAS15 reference over the 0.5-year window,
- SIMON-noNN over the same 0.5-year window.

The same rejection logic is applied:

- IAS15 failure,
- IAS15 ejection,
- IAS15 minimum-distance failure,
- IAS15 energy failure,
- noNN failure,
- noNN ejection,
- noNN energy failure,
- residual outlier,
- non-finite values.

Step 7: Save the same fields and summaries

v2 saves the same important arrays:

- full initial state,
- IAS15 final state,
- noNN final state,
- residual position,
- residual velocity,
- `X_raw21`,
- `X_re118`,
- scalar event diagnostics,
- energy diagnostics,
- minimum-distance diagnostics,
- noNN zone counts.

It also writes a summary with:

- number accepted,
- number tried,
- pass rate,
- `r_pair` statistics,
- `v_rad_norm` statistics,
- `v_tan_norm` statistics,
- approach/recede counts,
- `min_r_ias`,
- `min_r_nonn`,
- energy drift,
- residual RMS,
- noNN zone counts,
- rejection counts.

What v2 achieved

v2 improved the data near the observed real event in scalar-feature space. It generated samples with the right kind of strong-approach geometry, especially around the difficult $v_{rad,norm} \approx -1.1$, $v_{tan,norm} \approx 0.85$, $r \approx 0.139$ region.

However, v2 did not fully solve the deployed-event failure. The reason was that matching scalar pair features was not enough. The event could match in r , v_{rad} , and v_{tan} , while still being different in the full 18-dimensional state geometry.

That finding led to the nearest-neighbour diagnosis and then to v3 rollout-local data generation.

3.

`generate_encounter_surrogate_data_v3_rollout_local.py`

Purpose

`generate_encounter_surrogate_data_v3_rollout_local.py` is the final and most important data generator. It was created after diagnostics showed that the v2 event-targeted synthetic data still did not place the real rollout event close enough to the training set in full `X_re118` space.

The purpose of v3 is to generate training windows directly around the actual IC1 rollout event, not merely around synthetic states with similar scalar features.

The file explicitly states that the synthetic event-targeted generator matched the failed rollout event in simple pair features, but the real event was still not close to the training set in full `X_re118` state space. Therefore, v3 samples directly from the revised noNN IC1 rollout trajectory near the actual encounter time.

Step-by-step process

Step 1: Reuse already-tested noNN and IAS15 functions

Instead of reimplementing the integrators, v3 imports tested functions from `diagnose_single_event_replay_v1.py`, including:

- IC1 setup,
- noNN integration,
- IAS15 integration,
- pair velocity features,
- minimum-distance checks,
- energy checks,
- full `X_re118` construction.

This reduces the risk that v3 data is generated with a different definition from the replay diagnostic.

Step 2: Define the real-event local region

The default event region is:

```
t_lo = 82.20 yr
t_hi = 82.90 yr
event_time = 82.56 yr
dt = 0.08
window_years = 0.5
active pair = 1-2
```

The default anchor spacing is 0.02 years. This means the generator samples possible anchor states from the noNN trajectory around the encounter instead of inventing them synthetically.

Step 3: Build an anchor pool from the noNN rollout

The function `make_anchor_pool` starts from IC1, runs the revised noNN rollout up to the selected event-time region, and stores noNN states at multiple anchor times between `t_lo` and `t_hi`.

It also includes the exact event time if it lies inside the interval.

This creates the main difference from v1/v2:

- v1/v2 create synthetic states,
- v3 starts from actual noNN rollout states.

Step 4: Choose an anchor state

For each candidate sample, v3 chooses one anchor from the anchor pool. The time selection is biased toward the real event time using `time_sigma`, so samples are concentrated near the important encounter region but still spread over a local time interval.

Step 5: Apply small local full-state perturbations

The function `perturb_state` adds small perturbations to the full position and velocity state. These perturbations are deliberately small and centre-of-mass projected.

The perturbation has two parts:

1. full-state Gaussian perturbation, applied to positions and velocities;
2. optional pair-specific radial/tangential perturbation, applied to the active pair.

The code keeps the z -coordinates at zero for this prototype, matching the planar IC1 setup.

The purpose is to create local full-state coverage around the real event, without losing the true third-body geometry from the actual rollout.

Step 6: Re-centre the perturbed state

After perturbation, the state is shifted back to the centre-of-mass frame. This prevents artificial centre-of-mass drift from becoming part of the learned residual.

Step 7: Apply event filters

v3 requires the perturbed state to remain close to the intended event type. It checks:

- active pair 1-2 is in Zone 3,
- radial velocity is in the desired approaching range,
- tangential velocity is in the desired range,
- the state remains finite,
- the minimum distance behaviour is in the targeted range.

The default v3 settings target a strong-approach event around the actual IC1 encounter. In your local data-generation runs, the target filters included:

```
vr in roughly [-1.45, -0.40]
vt in roughly [0.50, 1.25]
target IAS15 min_r roughly [0.052, 0.085] AU
```

Step 8: Run IAS15 over the 0.5-year window

From the accepted perturbed start state, v3 runs IAS15 over the 0.5-year local window. This gives the reference exit state.

It rejects cases with:

- IAS15 failure,
- ejection,
- minimum-distance outside the target range,
- poor IAS15 energy,
- non-finite values.

Step 9: Run revised noNN over the same window

It then runs the revised SIMON-noNN window over the same 0.5-year endpoint.

It rejects cases with:

- noNN failure,
- noNN ejection,
- noNN energy problems,
- residual outliers,
- non-finite outputs.

Step 10: Compute and save the residual

As before, the target is:

$$residual_x = x_{IAS15,exit} - x_{noNN,exit}$$

$$residual_v = v_{IAS15,exit} - v_{noNN,exit}$$

The file saves the same core fields as v1/v2 so that the existing merge, inspect, and train scripts can be reused.

Step 11: Save sidecar diagnostics

Unlike v1/v2, v3 also writes a local diagnostic CSV sidecar. This is why v3 batches often produce three files:

```
.npz
_summary.txt
_local_diag.csv
```

The `.npz` contains the training data arrays. The `_summary.txt` contains shard-level statistics. The `_local_diag.csv` records per-sample local diagnostics such as anchor time, perturbation size, radial/tangential perturbations, and event geometry.

What v3 achieved

v3 solved the full-state coverage problem. It produced samples close to the actual real rollout event in full 3-body state geometry. This was necessary because v2 had only matched the event in scalar pair features, not in the full neural-network input space.

The final merged enriched dataset trained the successful model:

```
encounter_surrogate_v3_rollout_local_velocitysafe.pt
```

The final training summary reports:

- data: `encounter_surrogate_v3_rollout_local_enriched.npz`,
- samples: 3370,
- train/validation/test: 2359 / 506 / 505,
- model: MLP with 18 inputs, hidden size 128, and 18 outputs,
- loss: SmoothL1/Huber on normalized residuals,
- position weight = 1.0,
- velocity weight = 1.0,
- centre-of-mass consistency penalties,
- best epoch = 2335.

The dataset had broad enough coverage for the final IC1 $dt = 0.08$ local-event deployment:

- `r_pair` from about 0.0569 to 0.1474 AU,
- `v_rad_norm` from about -1.396 to +1.664,
- `v_tan_norm` from about 0.094 to 1.798,
- strong approach = 1921 samples,
- weak/side = 677 samples,
- recede = 772 samples.

How the old scalar Zone-3 dataset motivated the sampling choice

The old scalar Zone-3 dataset motivated the encounterNN data generation in an important but limited way.

What the old scalar dataset showed

The old scalar dataset showed that the key pair-level variables for Zone-3 correction were:

```
r_AU  
log_mi
```

```
log_mj
log_dt
v_rad_norm
v_tan_norm
```

It also showed that the radial velocity sign matters. Approaching encounters and receding encounters are not equivalent. This is why the encounterNN data generation explicitly separated:

- strong approach,
- weak/side,
- recede.

The old dataset also supported the decision to start with $dt = 0.08$, because this was a meaningful large-timestep regime in the Zone-3 work.

What the old scalar dataset could not provide

The old scalar dataset could not be used directly for encounterNN training because it did not contain:

- full 3-body positions,
- full 3-body velocities,
- third-body geometry,
- centre-of-mass state,
- IAS15 full final state,
- SIMON-noNN full final state,
- 18D residual target.

Therefore, it could only motivate the sampling variables and ranges. It could not be the actual encounterNN dataset.

How v1/v2 used it in code

The v1 and v2 code files include `--scalar-data`, defaulting to:

```
encounter_data_zone3_v3_augmented.npz
```

If that file was present in the run folder or explicitly passed by path, the generator loaded it and sampled empirical values of r , masses, radial velocity, and tangential velocity.

If it was not present, the generator used fallback random sampling ranges.

So the accurate statement is:

The old scalar Zone-3 dataset motivated the feature choices and, when available, was used by v1/v2 as an optional empirical proposal distribution. It was not used as encounterNN training data. The actual encounterNN training data was newly generated as full 3-body state-transition data.

How v3 changed the logic

The v3 rollout-local generator no longer depends on the old scalar dataset. It uses the actual revised noNN IC1 rollout around the real event. This change was made because scalar-feature matching alone was insufficient. The final model needed training states close to the actual event in full `X_re118` geometry.

Therefore, the final data-generation story is:

1. The old scalar dataset motivated which encounter variables mattered.
2. v1 converted those ideas into a new full-state synthetic generator.
3. v2 targeted the difficult real-event scalar region.
4. v3 used actual noNN rollout-local states to fix full-state distribution mismatch.
5. The final training data was a newly generated full-state residual dataset, not the old scalar dataset.